Where do libraries come from, and where do they stop helping?

Bootstrapped libraries, documentation, evidence selection, and boundaries

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In one sentence

A 4-bit 8B model wrote a verified library for a new benchmark from three examples, and a 0.6B caller that never saw the benchmark uses it correctly to 1M tokens; the useful capacity is in the proposer and in choosing evidence, not in readers, recursion or larger controllers; a retrieval library also beats direct reading on LongBench-v2.

Libraries can be bootstrapped: an 8B wrote MRCR's library.

MRCR held out (8k to 1M tokens)	2 needles	4 needles
Direct answer, 4B	0.23	0.15
RLM, untrained 4B, no library	0.18	0.11
8B-written library, run alone	1.000	1.000
0.6B caller + library	**1.000**	**1.000**
Untrained 4B + library	0.95	0.88

64 rows per column. The 0.6B caller never saw MRCR. The proposer saw the contract and three solved examples; candidates run on 30 teaching tasks; the best scored 30/30 and was reviewed before use (40 lines of parsing, nothing copied from the examples).

Who can write it (seeds accepted)

4B Instruct: 0 of 1
4B, thinking off / on: 0 of 1 / 0 of 2
8B, thinking off: 0 of 2
8B, thinking on: **2 of 5**
14B (4-bit), thinking on: **2 of 5**
4B after SFT + RL: 0 of 4; test-time RL on MRCR: 0 in 1,920 samples

Caveat: MRCR is solvable by a short parser, so this measures program discovery, not reading. We found no published note saying so.

Ledger E21, E21b; runs e21-*. MRCR: openai/mrcr (MIT), official SequenceMatcher grader.

Discovery on 37 generated families: thinking, scale, and training.

Proposer	Solved / 37	Tokens
4B Instruct (2 seeds)	5, 4	4k
8B, no thinking	4	4k
4B, thinking	14	55k
8B, thinking (2 seeds)	20, 25	48k
14B, thinking (2 seeds)	21, 26	30k
4B + SFT (2 seeds)	9, 7	3k
4B + SFT + RL (2 seeds)	16, 14	2k
4B, round four (2 seeds)	**22, 20**	2k

Tokens: proposer output per family. Families vary record format (pipe, kv, csv, prose, XML-like, chat), question and answer format; held-out families use a new format or a new question.

Training the proposer: SFT on the 8B's verified code, then 40 steps of RL with the verified teaching score as reward: 9 to 30 of 74 attempts ($p < 0.001$); with more varied teacher data, 42: no different from the 8B (45) or 14B (47). Scale shows only where the helper is needed (families at 0.7+: 1, 5, 6 of 8). MRCR still fails.

Helper gap: 3x the helper teacher data brings the 4B to the 8B on helper families (mean 0.46), at about 6 code families (ns).

Ledger E26 to E26g, E36, E41; runs e26-disc-*, e41*. Generator: family_gen.py (answers exact).

Bootstrapping is cheap where answers are exact (pre-registered).

30 unseen families, 2 seeds	Correct	False acc.	Tokens
4B, best of 8, 3 labels	**12, 14**	0, 1	3.7k
8B thinking, 30 labels	13, 16	0, 0	52.4k

Correct = accepted library scores 0.9+ on 30 fresh tasks with twice the records. Pooled 26 vs 29 of 60, p 0.61. Protocol, predictions and analysis committed before the runs; all predictions met in both seeds.

Why three labels suffice

Candidates that run cleanly and reproduce the prompt's examples are correct: precision 0.81, 0.88, 0.98 with 1, 2, 3 labels (127 of 129 at three, all generalizing). Agreement among candidates is no substitute: an RL-trained proposer repeats its wrong programs.

Limit: on noisy families (MuSiQue F1) ranking close routines takes tens of labels.

Ledger E37, E38, E39; runs e37*, e39*; docs/notebook/assets-e39-preregistration.md.

One line of documentation decides whether an untrained caller uses a library.

MRCR, untrained 4B caller, docstring	Score
Signature only	0.18
Description + signature (as the 8B wrote it)	0.59
Signature + usage line	0.95
Description + signature + usage line	0.95
Hand-written library, usage line removed	0.22
Hand-written library, as written	0.95

Usage line: `Typical use: FINAL_TEXT(answer_mrcr(request)).`

Without it the untrained 4B hands the whole input to a sub-model (overflowing at 128k and above) or retypes the routine's output by hand.

The distilled 0.6B scores 1.000 without the line; the untrained 0.6B fails either way (0.21, 0.13). Robustness comes from distillation, not size. The bootstrapper now inserts the line.

Ledger E25c, E25d; runs e25c-*, e25d-*. One family, one library, one seed per arm.

MuSiQue in long haystacks: capacity belongs in choosing evidence.

Answer F1 (90 to 180 held-out questions per row)	8k	32k	128k	tokens
Direct answer, 4B	0.52	0.25	overflow	
RLM without a library, depth 0 / depth 1	0.49 / 0.51	0.25 / 0.25	overflow	
Library that hands each hop to a child RLM	0.29	0.29	0.34	3 to 4 times
0.6B caller + filter, BM25, read	0.52	0.31	0.46	57k
0.6B caller + two-round retrieval	**0.53**	**0.39**	**0.50**	10k

What helps: a second, targeted retrieval (BM25 for the question, the helper names the missing fact, BM25 again, one read): +0.058 over BM25 alone, +0.045 over filter+BM25 at a sixth of the tokens.

What does not: recursion; an 8B reader (+0.01; gold paragraphs reach 0.63 to 0.67 with either reader); more sampling (majority of 3 at 88k tokens: 0.22 vs 0.35 for the library at 32k).

Ledger E22c to E22f, E23, E25e, E28. MuSiQue (CC BY 4.0) dev questions in haystacks of other questions' paragraphs.

External validity, and what we withdrew.

LongBench-v2 (all 503 questions, chance 0.25)

Direct (truncated): 0.356, 42k tokens
Retrieve-then-read: 0.427, 10k tokens
via the 0.6B caller: 0.423 (+0.07, se 0.02)

Selecting evidence beats summarizing (map-reduce: no gain).

Boundary: no proposer bootstraps OOLONG, our own family (8B 0.20, 14B 0.00; shipped library 0.60): 30 examples span 11 question templates.

Withdrawn: distractor-limited reader; length routing (pre-registered); "no help on LongBench-v2" (it was map-reduce).

Method lessons

- Verify under the semantics you run under: the kernel's `ask` raises on a cut-off reply, scripted harnesses returned partial text (one routine crashed in 46 of 90 caller runs).
- Verification must span deployment lengths: 8k-only teaching selected a pass-through routine that scores 0 at 128k.
- Routine evolution (GEPA-style): prompt rewrites by the 8B hurt; numbers helped (+0.048, ns); random search found nothing.

Ledger E32 to E32e, E31, E22b, E24, E24c, E35. LongBench-v2: zai-org (Apache-2.0).

What this changes, and what to test next.

The thesis, sharpened. For small recursive models, capability lives outside the weights: verified routines (bootstrapped by a capable proposer, or by a small one trained on verified code and rewarded by verification; on exact families three labels suffice), documentation a thin caller can act on, and evidence selection inside the routine. A 0.6B caller with a retrieval library beats a 4B reading the document on MuSiQue and on LongBench-v2.

Next

1. Helper gap: more of the same teacher data stops helping (held-out code falls 42, 34, 28 of 58 as helper pairs grow 20, 65, 113); next try more varied helper questions per pair, not more pairs, with two RL seeds per recipe.
2. Labels on noisy families: graded verification for MuSiQue- and OOLONG-like families.
3. Breadth: split a broad family into question templates automatically (OOLONG: counting yes, timeline and user no).
4. A short note on program-solvable long-context benchmarks (MRCR; candidates: Michelangelo, RULER).

Appendix: protocol and provenance.

Models: Qwen3-4B-Instruct-2507 (helper, caller), Qwen3-0.6B + LoRA (distilled caller), Qwen3-4B hybrid and Qwen3-8B-AWQ (proposers), all served with vLLM 0.27.1 on two A100 MIG 7g.40gb slices.

Data: OpenAI MRCR (MIT; within-bin split, 30 teaching rows, 64 held out per needle count); MuSiQue v1.0 (CC BY 4.0; teaching from train, held out from dev; two samples of 90); LongBench-v2 (Apache-2.0, revision 2b48e494); generated families (family_gen.py, exact answers).

Statistics: paired differences with standard errors over all questions; exact McNemar for family counts; the routing test was pre-registered.

Every number above has a ledger row in docs/RESULTS.md and a run directory in the run store; the lab notebook (docs/notebook/2026-10-19-...) records failures and corrections in time order.